Runsheng Chen

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Education

Technical University of Munich (TUM) — Munich, Germany
Ph.D. in Computer Science, Legal Tech Group (Advisor: Prof. Matthias Grabmair) — *Jun 2025 – Dec 2028 (Expected)*

Technical University of Munich (TUM) — Munich, Germany
M.Sc. in Robotics, Cognition, Intelligence; Grade: 1.8, Thesis: 1.0 (Top Grade, in collab. w/ Google DeepMind) — *Oct 2021 – Mar 2025*

Southeast University — Nanjing, China
B.Eng. in Mechanical Engineering — *Sep 2016 – Jun 2020*

Publications

- Mohamed Hesham Elganayni*, **Runsheng Chen***, Sebastian Nagl, Matthias Grabmair. *Exploiting LLM-as-a-Judge Disposition on Free Text Legal QA via Prompt Optimization.* **ICAIL 2026**. [arXiv:2604.20726] (*equal contribution)
- Niklas Wais, Max Prior, **Runsheng Chen**. *AppeaLLM: Mimicking a Closed-World Environment for Court Decision Prediction.* **ACM CS&LAW 2026**. [doi:10.1145/3788646.3789528]
- Yihong Liu*, **Runsheng Chen***, Lea Hirlimann, Ahmad Dawar Hakimi, Mingyang Wang, Amir Hossein Kargaran, Sascha Rothe, François Yvon, Hinrich Schütze. *On Relation-Specific Neurons in Large Language Models.* **EMNLP 2025 Main** (Master's Thesis, in collab. with **Google DeepMind**). [arXiv:2502.17355] (*equal contribution)

Research Experience

Generatives Sprachmodell Justiz (GSJ) — Legal Tech Group, TUM
LLM Post-Training, Alignment & Legal NLP Systems — *Jun 2025 – Present*

- Collaborated in a project team for the German Justice Ministry, formulating post-training **recipes** and engineering scalable **distributed pipelines** using **RL** and **on-policy distillation (OPD/OPSD)** for legal reasoning and summarization.
- Engineered the **agent runtime harness** wrapper around the legal agent, managing context window lifecycle, tool-use state machines, sandboxed execution, and multi-turn workflow integration.

On-Policy Self-Distillation on Legal Reasoning Benchmarks — Legal Tech Group, TUM
LLM Post-Training & Knowledge Distillation — *May 2026 – Jul 2026*

- Formulated on-policy self-distillation (OPSD) workflows for compact LLMs (**Qwen3-4B**) on the LEXAM reasoning benchmark to mitigate exposure bias and multi-step reasoning degeneration.
- Systematically evaluated the role of **privileged information** (gold rationales, statutory hints, step-level teacher critiques) in stabilizing policy distillation trajectories and boosting OOD generalization.

Answer-Aware Hint Refinement for RL-Trained Search Agents — Legal Tech Group, TUM
Agentic RAG, Reinforcement Learning & Tool-Use — *Mar 2026 – May 2026*

- Advanced agentic search training (extending Search-R1/R2 on **Qwen2.5-3B/Qwen3-8B**) by proposing **answer-aware, loss-masked hint refinement** to alleviate credit assignment and reward sparsity in multi-hop query trajectories.
- Engineered a distributed RL training infrastructure across an **8×H200 GPU cluster** using **VeRL, vLLM**, GPU-accelerated FAISS indexing, and an asynchronous LLM-judge reward server.

Open Source & Research Tools

Zotero Smart Highlighter — github.com/MemorushB/zotero-smart-highlighter
Intelligent Document Assistant Plugin for Zotero 8 (TypeScript, Python) — *Open Source*

- Developed and released an intelligent document-reading tool that extracts and highlights key informative passages in research papers across three configurable backends: local **BM25 ranking**, **on-device neural reranker**, and LLM APIs.

Industry & Engineering Experience

Accenture — Munich, Germany
Working Student — AI Automation & Enterprise Workflows — *Sep 2024 – Apr 2025*

BMW Group — Munich, Germany
Machine Learning Engineering Intern — Operational AI in Production — *Nov 2023 – May 2024*

Technical University of Munich (TUM) — Munich, Germany
Graduate Research Assistant (HiWi) — Educational AI Systems — *Aug 2023 – Jul 2024*

Technical Skills

- **Research Areas**: Large Language Models (LLMs), Post-Training (SFT, RL / PPO / GRPO, On-Policy Distillation), Agentic Search / Tool-Use RAG, Mechanistic Interpretability, Evaluation Benchmarks
  Frameworks & Systems: PyTorch, Hugging Face (Transformers, TRL), VeRL, vLLM, SGLang, DeepSpeed, Ray, FAISS, Docker, Git
  Languages & Tools: Python (Proficient), C++, R, Bash, SQL, LaTeX | English (C2/Bilingual), German (C1), Chinese (Native)